Hence,

$$0 = u^2 \sin^2\theta - 2gh_{max}$$

$$\text{Or } h_{max} = \frac{u^2 \sin^2\theta}{2g} \qquad (2.32)$$

Time of flight (T_f)

The total time taken by the projectile from the point of projection till it hits the horizontal plane is called time of flight.

This time of flight is the time taken by the projectile to go from point O to B via point A (Figure 2.43)

We know that $s_y = u_y t + \frac{1}{2} a_y t^2$

Here, $s_y = y = 0$ (net displacement in y-direction is zero), $u_y = u\,sin\theta$, $a_y = -g$, $t = T_f$ Then

$$0 = u \sin\theta\, T_f - \frac{1}{2} g T_f^2$$

$$T_f = 2u \frac{\sin\theta}{g}$$

Horizontal range (R)

The maximum horizontal distance between the point of projection and the point on the horizontal plane where the projectile hits the ground is called horizontal range (R). This is found easily since the horizontal component of initial velocity remains the same. We can write

Range R = Horizontal component of velocity x time of flight = $u\cos\theta \times T_f$

$$R = u\cos\theta \times \frac{2u\sin\theta}{g} = \frac{2u^2 \sin\theta\cos\theta}{g}$$

$$\therefore R = \frac{u^2 \sin 2\theta}{g} \qquad (2.33)$$

The horizontal range directly depends on the initial speed (u) and the sine of angle of projection (θ). It inversely depends on acceleration due to gravity 'g'

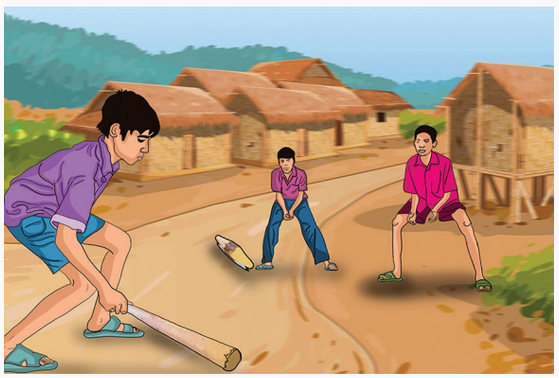
For a given initial speed u, the maximum possible range is reached when $\sin 2\theta$ is maximum, $\sin 2\theta = 1$. This implies $2\theta = \pi/2$

or $$\theta = \frac{\pi}{4}$$

This means that if the particle is projected at 45 degrees with respect to horizontal, it attains maximum range, given by.

$$R_{max} = \frac{u^2}{g} \qquad (2.34)$$

Projectile motion !



In Tamil Nadu there is an interesting traditional game 'kitti pull'. When the 'pull' is hit by the kitti, the path followed by the pull is 'parabolic'.

2X5428

UNIT-2(XI-Physics_Vol-1)-1.indd 86 28-01-2020 17:56:47